

DOSY

How to address its difficulties

A practical walkthrough · 20 minutes

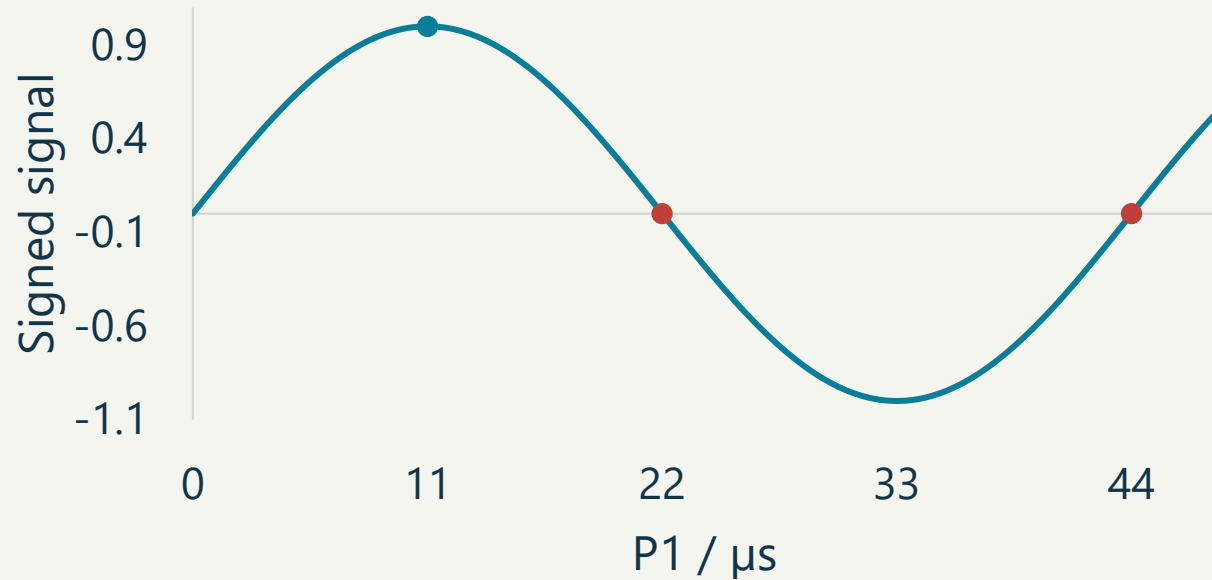
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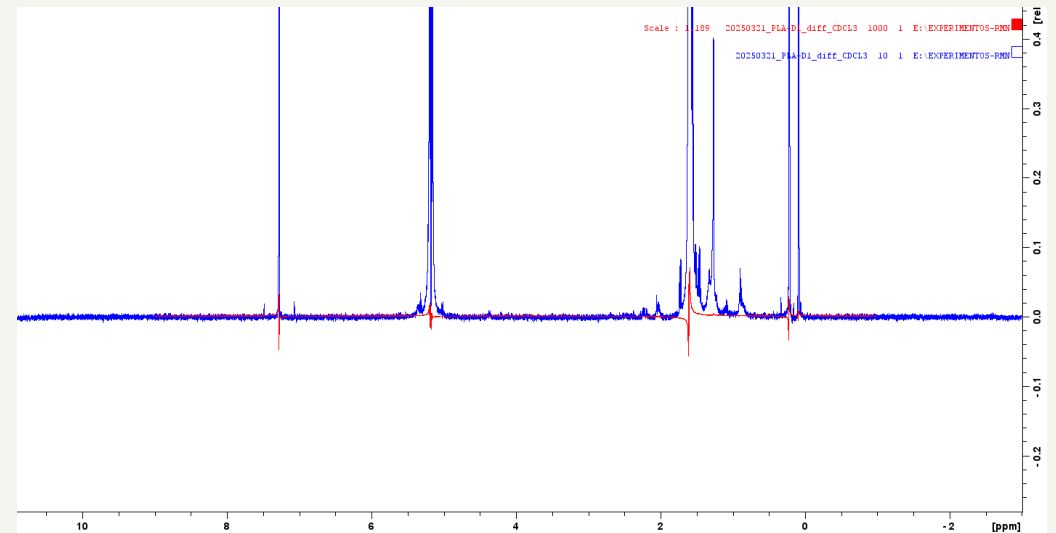
Measuring, processing and checking DOSY

- 01** **PREPARE THE MEASUREMENT** Calibration and parameter selection
- 02** **UNDERSTAND THE DATA** Processing and inversion can be misleading
- 03** **USE THE TOOLS** Our software and practical cases
- 04** **CHECK THE RESULT** Documented successes and the next ultrafast step

First difficulty: is this really a 90° pulse?



— Ideal zg model ● 90° ● 180° / 360°

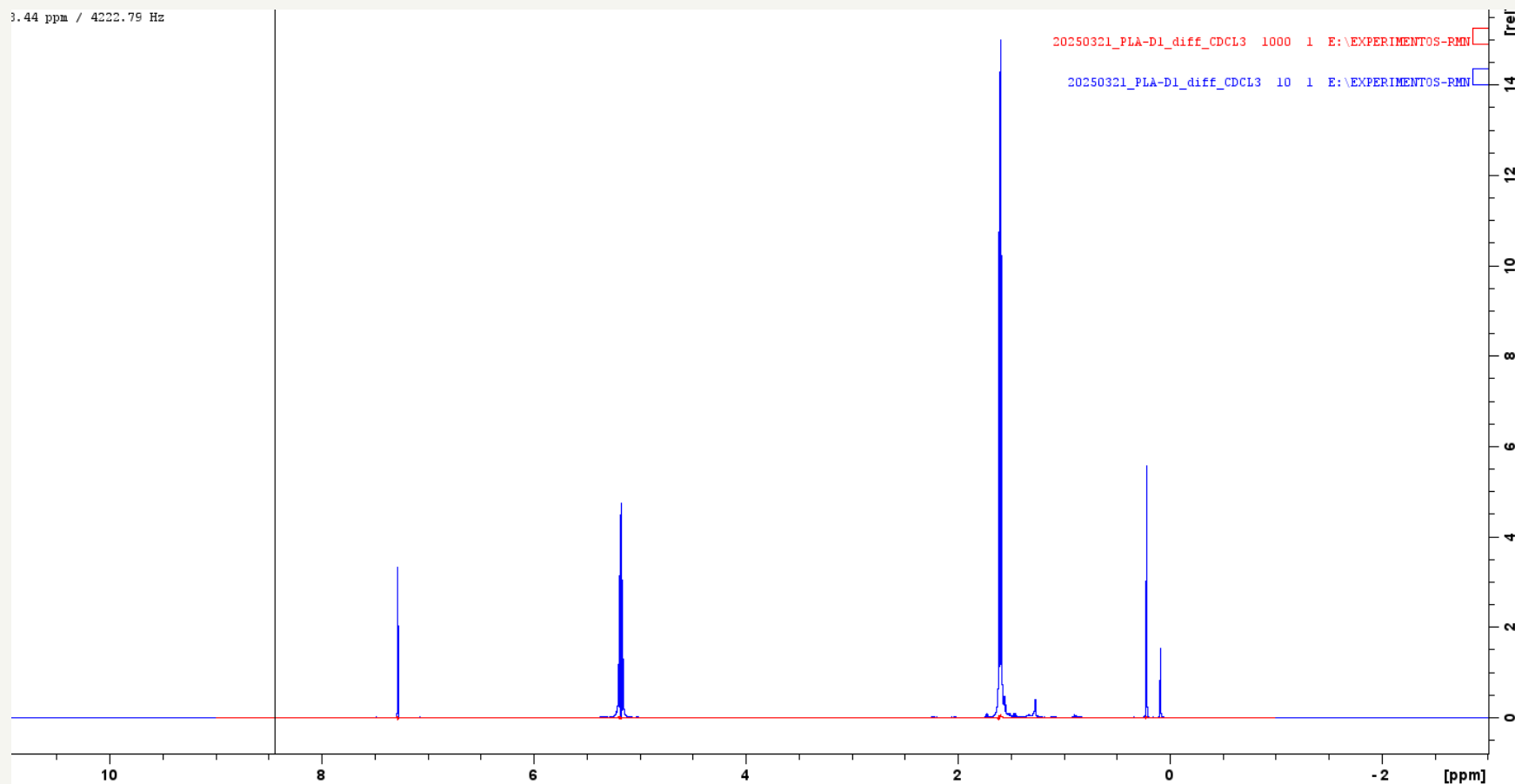


Fixed phase and power

p1 → zg → efp · fixed phase · first null → $P1 / 2$ → verify

Confusing 180° with 360° changes P90. Use sufficient D1 and fixed RG; inspect artefacts and coherences.

Case 1 · from 22 μs to P90 = 11 μs



EXPNO 1000 · zg

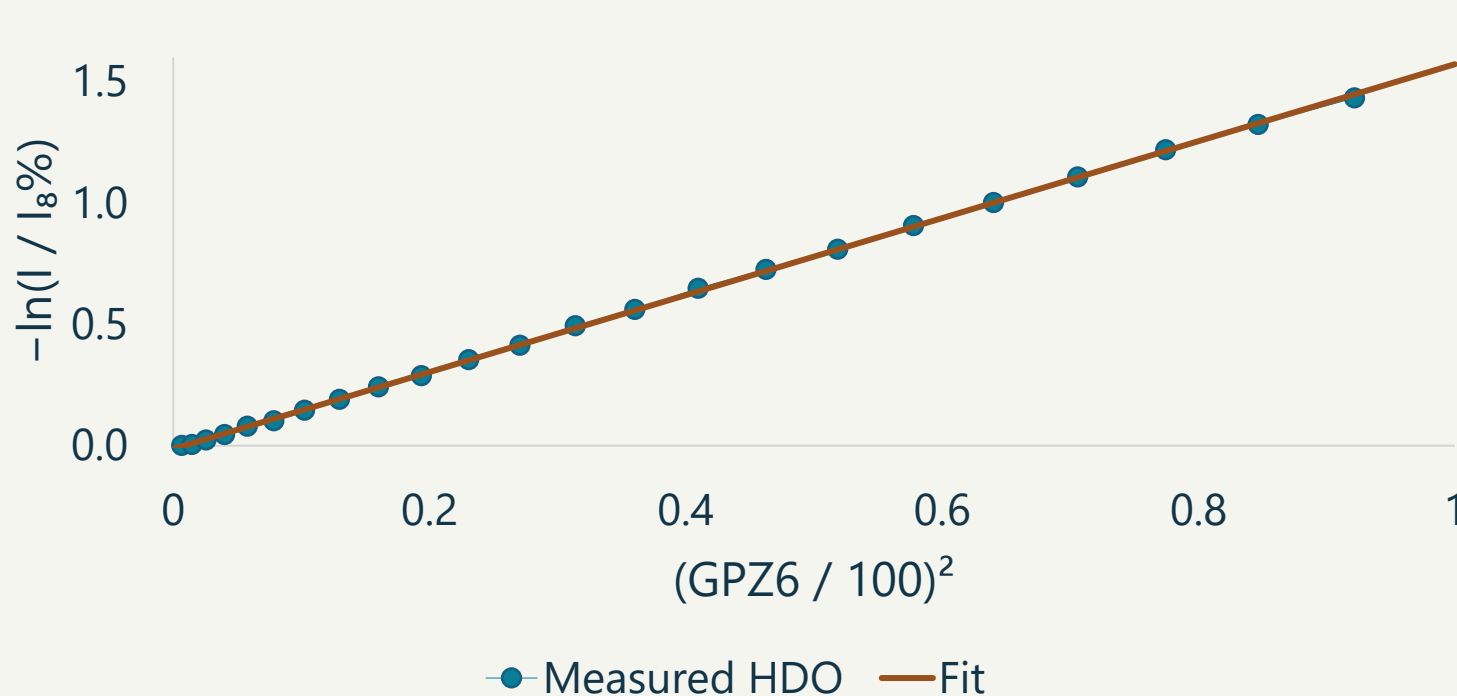
22 μs

$\div 2$

P90 = 11 μs

User-identified 180° null. Blue: 90°. Different RG/NS in the screenshots: qualitative comparison.

Gradients: calibrate a scale that enters quadratically



HYPOTHETICAL ERROR EXAMPLE

G actual = 1.10 G used

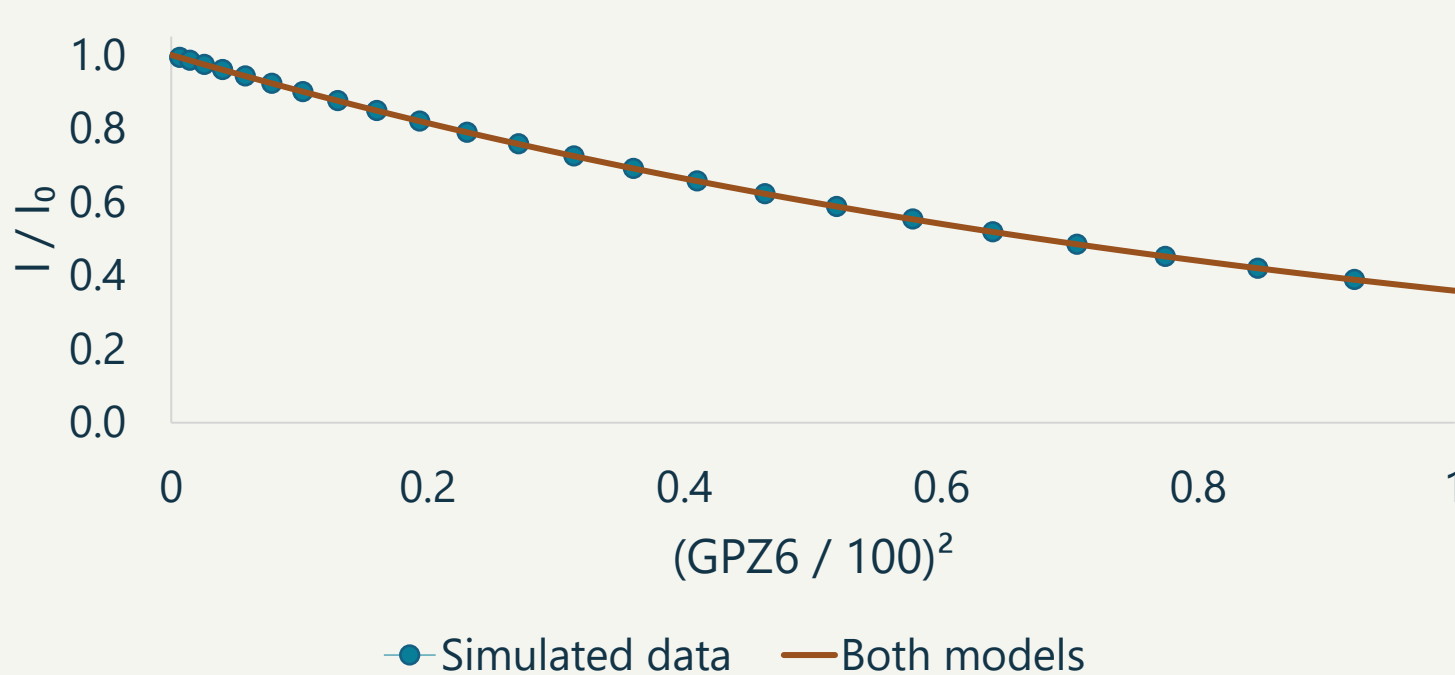
Apparent D = 1.21 D

HDO / TBO
|slope| = 1.58627

Standard at actual T · fixed region · residuals · check a second ramp

$G_{\text{new}} = G_{\text{previous}} \sqrt{(D_{\text{measured}} / D_{\text{ref}})}$. TBO profile: $\Delta = 50$ ms, $\delta = 1.20$ ms, SMSQ10.100.

The sequence determines what b means



$$R^2 = 1$$

$$D / 10^{-9} \text{ m}^2 \text{ s}^{-1}$$

SMSQ / bipolar: 1.000

PGSE: 0.802

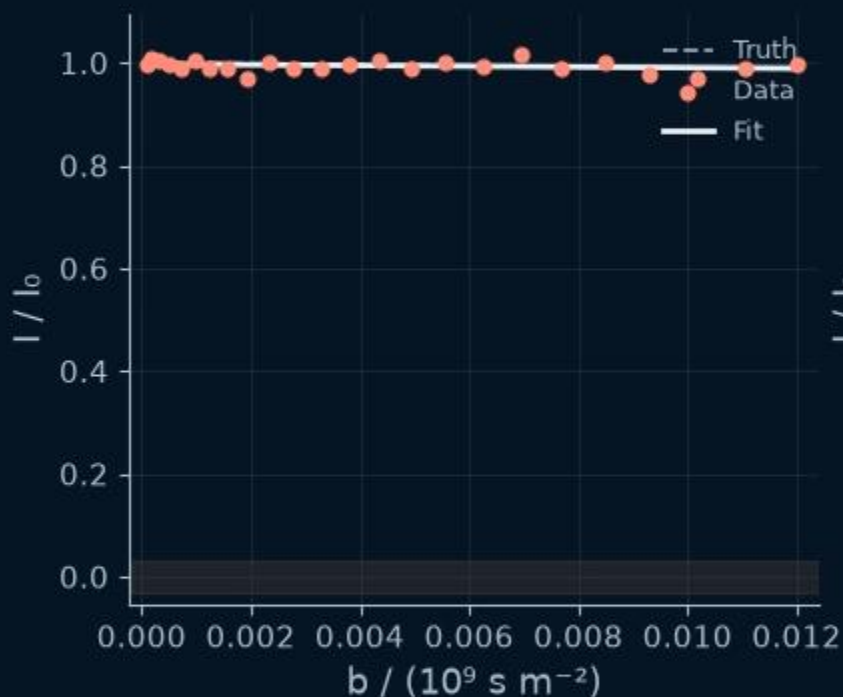
Inspect sequence + shape + δ + RF gaps before fitting D

NOISELESS SIMULATION, fixed physical G and no recalibration. The fit checks the decay, not the b scale.

The ramp can prevent a reliable D estimate

Same species, same absolute $\sigma = 1\% I_0$; 24 points; true $D = 1.000 \times 10^{-9} \text{ m}^2/\text{s}$

Almost flat

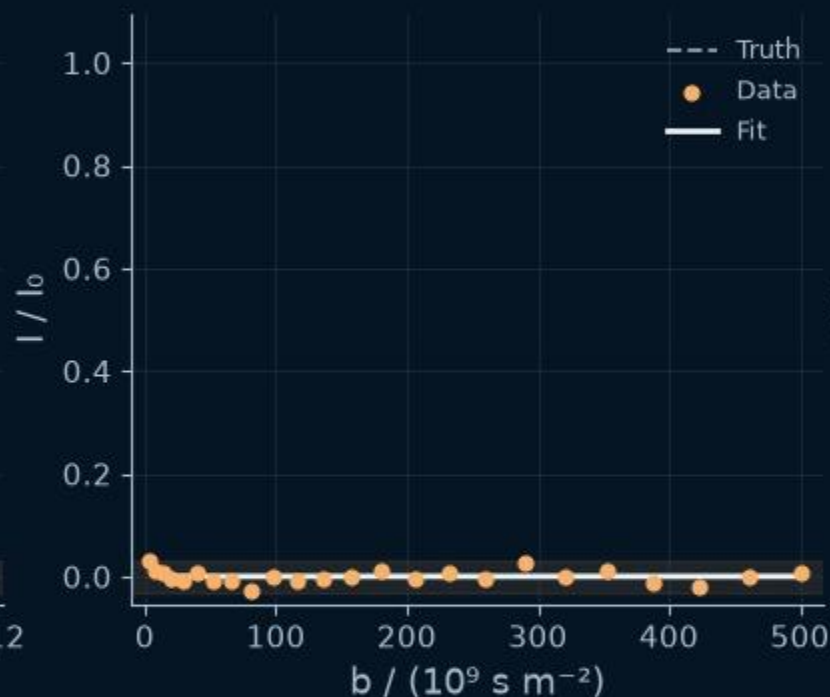


$D = 0.714$ CI 95% [0.00, 1.81]

Residual RMS = 1.06σ

Signal $\geq 3\sigma$: 24/24

Almost extinguished

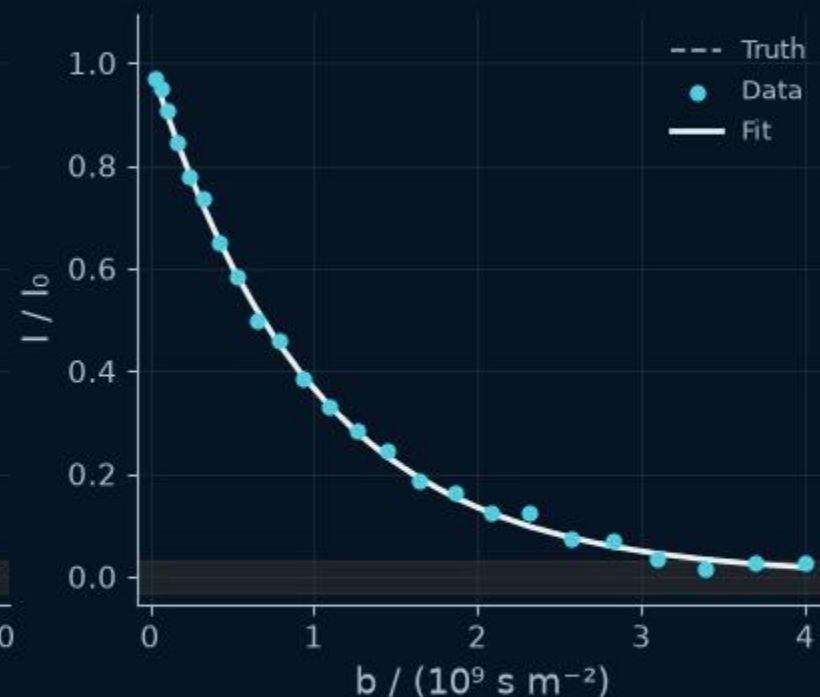


$D = 0.237$ CI 95% [0.05, ∞]

Residual RMS = 1.05σ

Signal $\geq 3\sigma$: 1/24

Useful range



$D = 1.007$ CI 95% [0.99, 1.03]

Residual RMS = 1.06σ

Signal $\geq 3\sigma$: 22/24

P30 × D20

P30: the effect grows quadratically

SIMULATION · fixed absolute noise



Same noise; relaxation reduces the available signal.

$D = 0.8 \times 10^{-9} \text{ m}^2/\text{s}$ · $G_{\text{max}} = 0.5 \text{ T/m}$ (assumed) · $\delta = 2P30$

P30

2.63 ms

D20

50 ms

SNR start → end

238 → 0.0

Final / initial signal: < 0.1%

$b_{\text{max}} = 17.82 \times 10^9 \text{ s/m}^2$

Tail reaches the noise:
many points add little.

LED and double STE: recover signal without copying delays

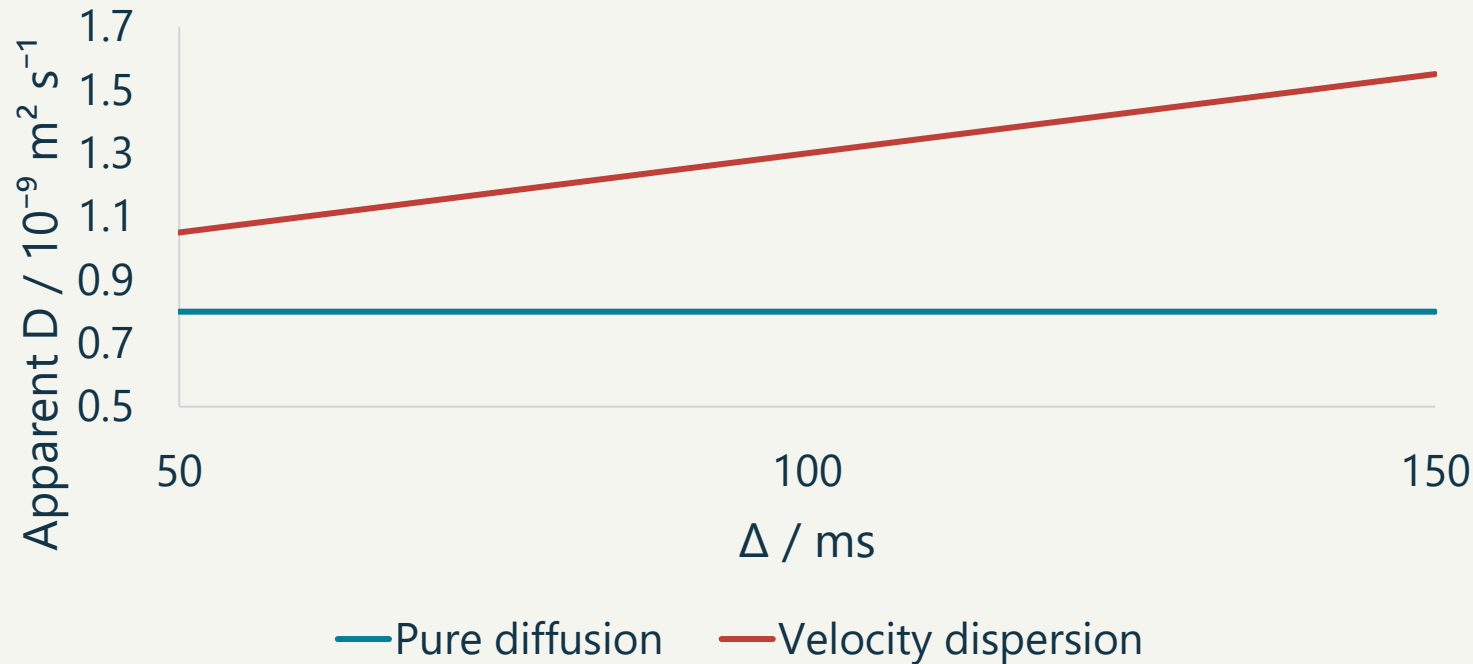
Sequence	Relevant delay	Problem to control
ledbp2s / 1d	LED = D21	Eddy-current recovery
dstebp3s / 1d	D20 total · D21 LED	Convection / more RF / sensitivity loss
diffSteLED	D5 ≠ LED · LED = D19	D5: remaining diffusion interval

Double STE: $\delta = 2P30 \cdot 8 \text{ lobes} \cdot NS = 16 \times n$

Read the acquired pulseprogram and its constraints before copying parameters

Inspect D1, D16, P19 and DELTA1. Convection compensation has limits and a signal cost.

Changing Δ helps reveal that diffusion is not the only process



$\Delta \uparrow$
Apparent $D \uparrow$

Convection
Exchange
Restricted diffusion

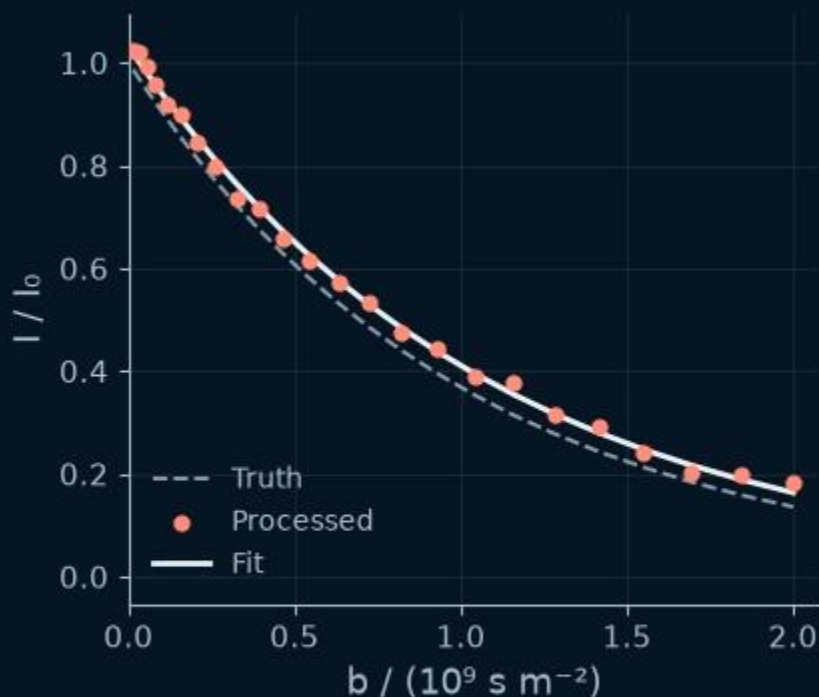
Compare three Δ values, T and ramp-specific amplitudes; check the standard

SIMULATION: Gaussian velocities, $\sigma v = 100 \text{ } \mu\text{m/s}$, narrow pulses. The cause needs additional evidence.

Processing can manufacture a convincing D

True $D = 1.000 \times 10^{-9} \text{ m}^2/\text{s} \cdot \sigma = 1\% I_0$ · The fitted model remains $A \exp(-bD)$

Additive offset +4% I_0

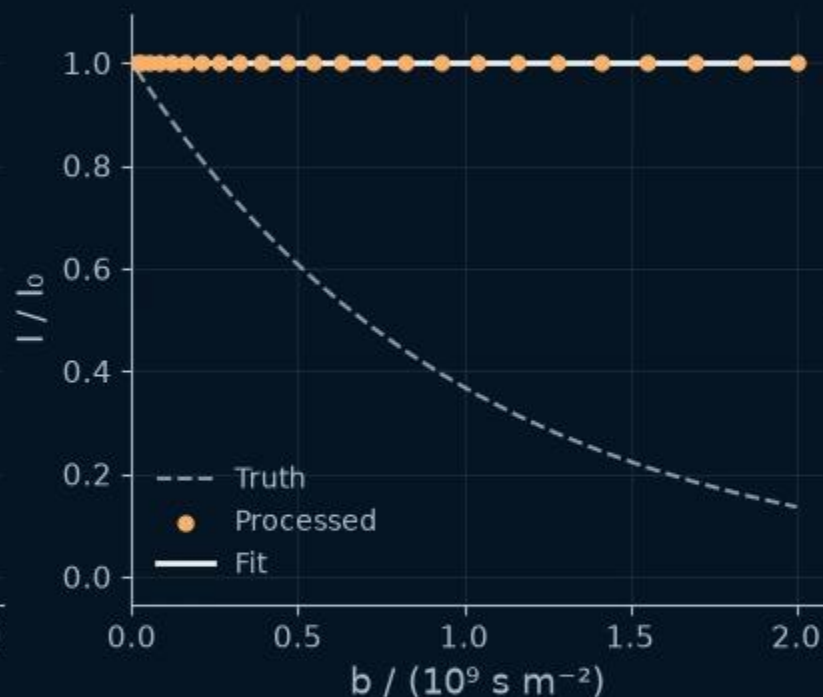


$D = 0.922$ CI 95% [0.90, 0.94]

$\text{RMS} = 1.20 \sigma$

Bias despite small residuals

Each point / itself

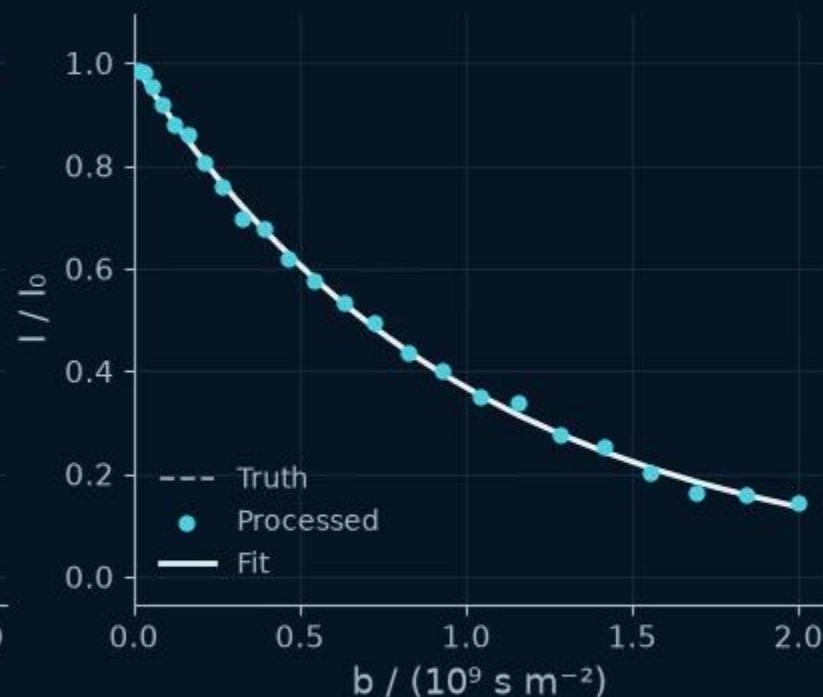


$D = 0$ RMS = 0

$I_i / I_i = 1$

Attenuation destroyed

Subtract a measured offset



$D = 0.999$ CI 95% [0.98, 1.02]

$\text{RMS} = 1.06 \sigma$

Offset known in this simulation

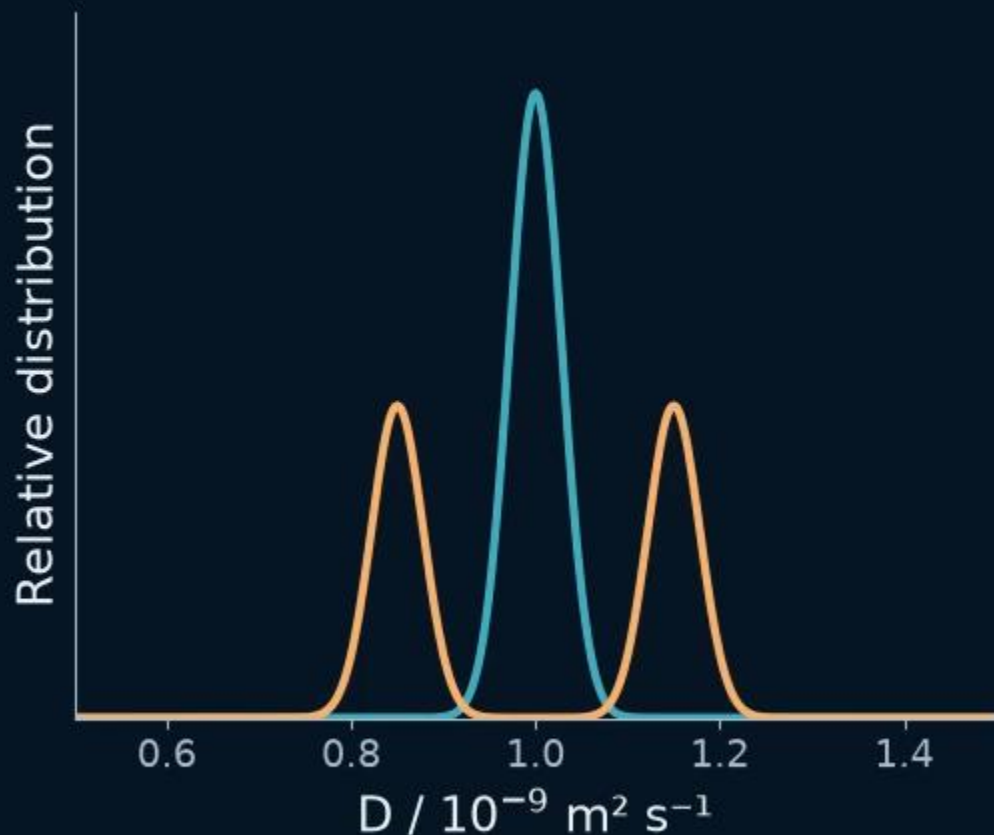
SIMULATION · Keep phase, integration window, scale and reference consistent across the series

Different distributions, nearly identical curves

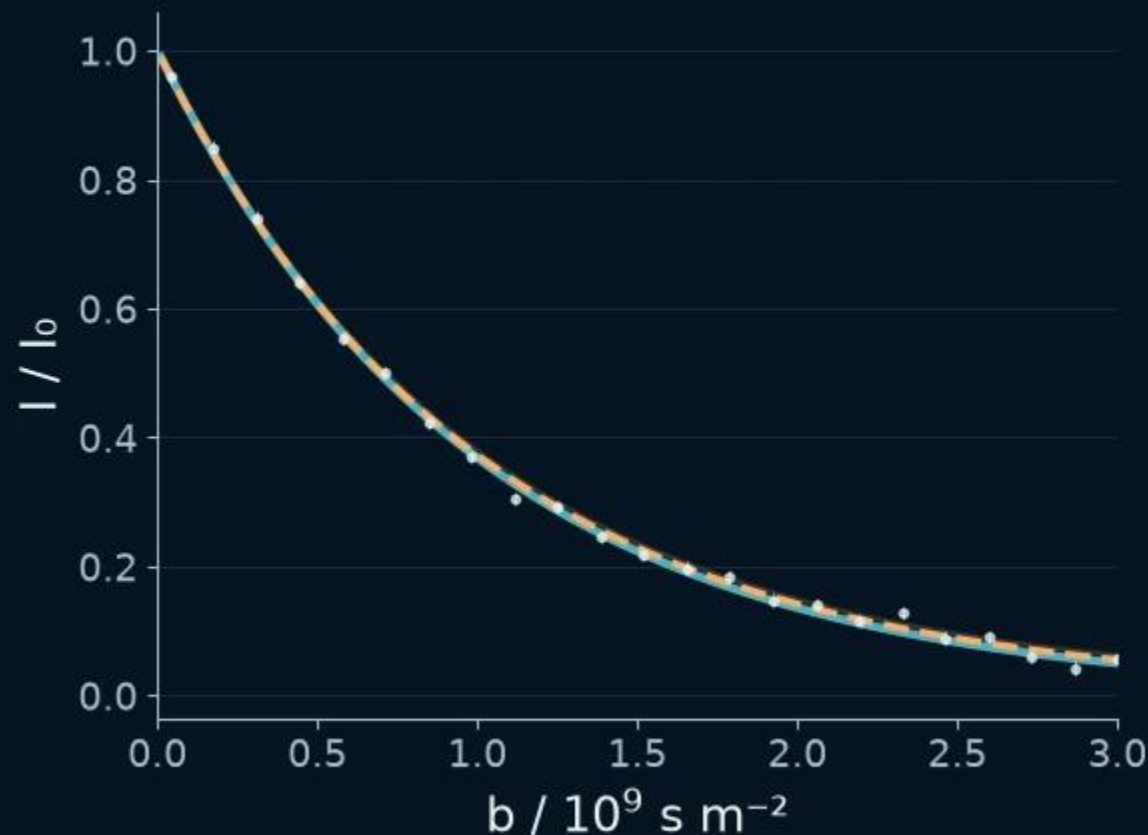
Signal uncertainty limits what we can resolve.

One component

Two components



Maximum difference: 0.61% of I_0

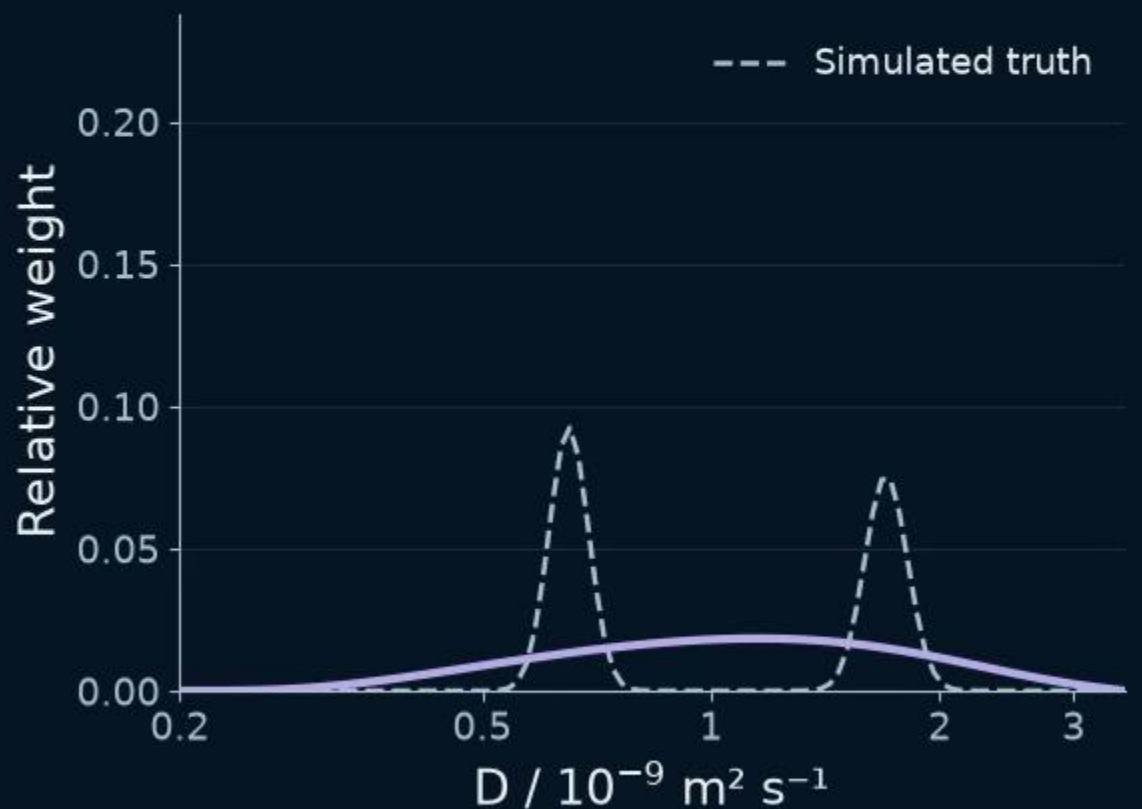


SIMULATION · noise $\sigma = 1\%$ of I_0 · not a measurement

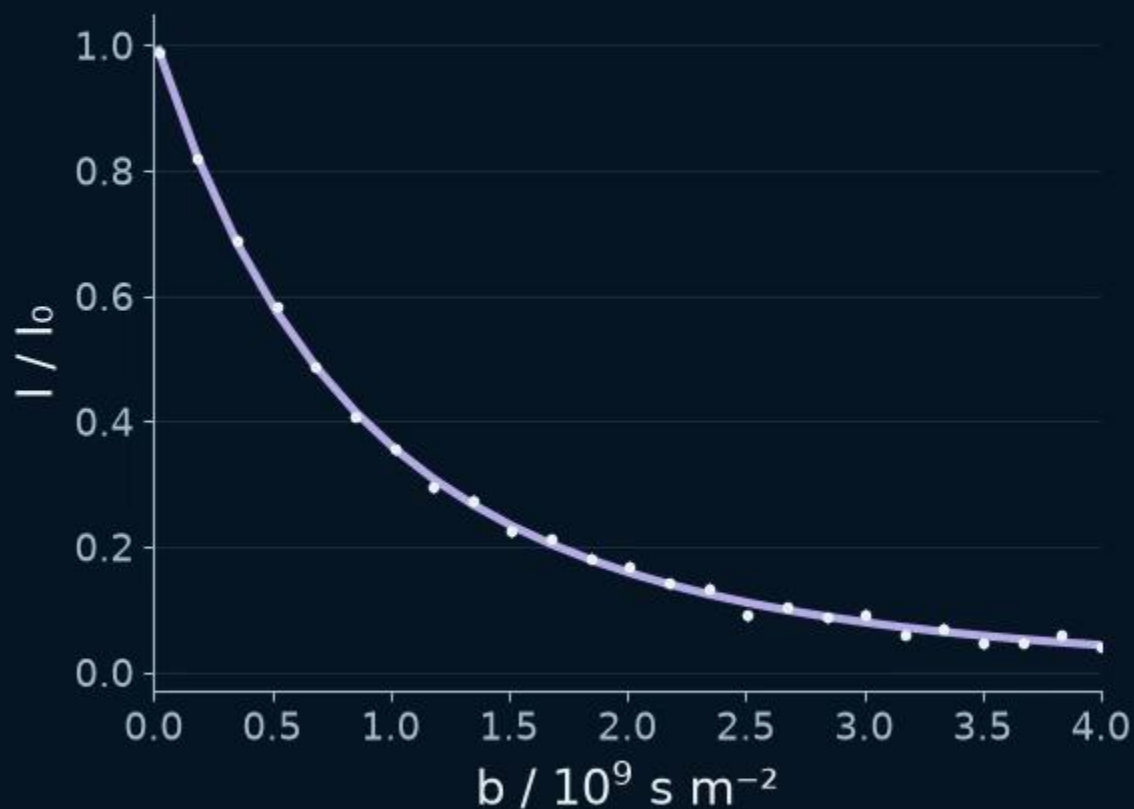
Regularization changes the solution

Same data and same grid. Change the smoothness weight.

Strong regularization $\lambda = 200$



Smoothing can merge components

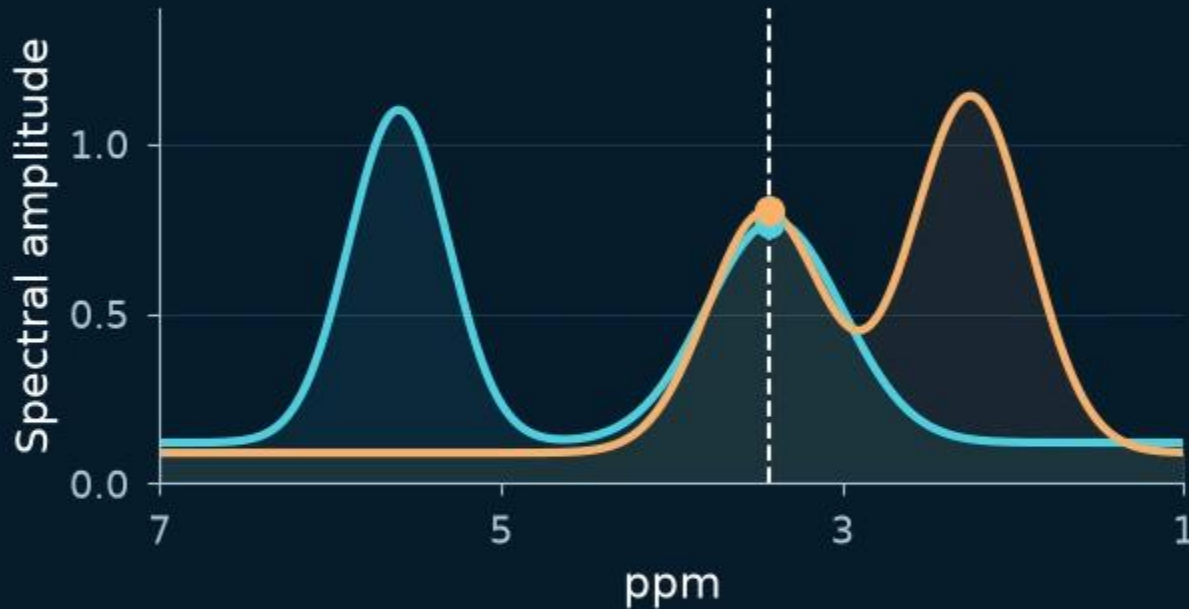


SIMULATION · NNLS + L_2 second-difference penalty · λ depends on this scaling

The whole spectrum shares information

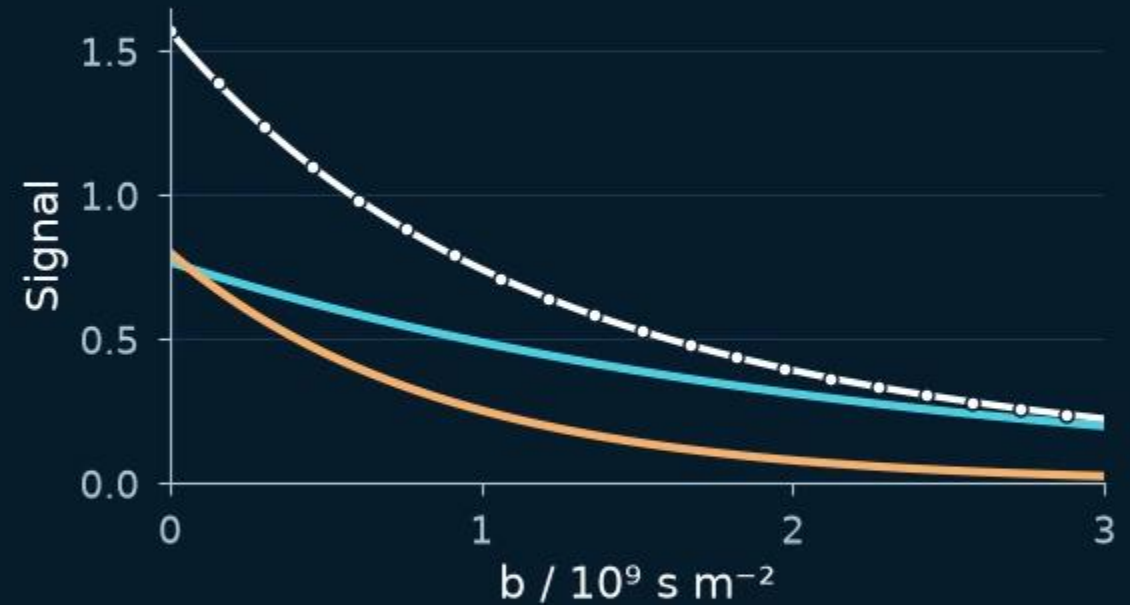
Two diffusion profiles; different amplitudes at each frequency.

AMPLITUDES $A_1(\nu)$, $A_2(\nu)$



$\nu = 3.43$ ppm

DECAY AT THE SELECTED ν



$$D_1 = 0.45 \times 10^{-9}$$

$$D_2 = 1.15 \times 10^{-9} \text{ m}^2/\text{s}$$

$$S(b, \nu) = A_1(\nu)e^{-bD_1} + A_2(\nu)e^{-bD_2}$$

The work our programs help us handle

**We know what needs to be checked.
Now we put it into practice with our tools:
prepare, analyse and preserve the result.**

Ramp and calibration

One signal and the whole spectrum

Sample series and controls

These are the problems we developed our tools to address.

Our programs and their functions

Tool	Input	What it provides
TopSpin · xpy	Template and 1D series	Prepare ramps and integrate the standard
DiffAtOnce Prime	Processed Bruker series	Local decay, map and diagnostics
ResinAtOnce	Resin / polymer series	Guided workflow, references and export
DALTAIL · MATLAB	Data and saved configuration	Process batches and reproduce tables

The next cases show the input, the operations and the result.

TopSpin: an HDO calibration report

xpy dosy_workshop.py

- 1 Select the series and a fixed region
- 2 Integrate and provide Dref(T)
- 3 Review the fit and save the proposal

REAL HDO · OFFLINE REPLAY

$R^2 = 0.999775$

$\Delta TE = 0.77 \text{ K}$

Configured limit: 0.50 K

The program detects drift: stabilize and repeat before calibrating

Offline replay. Illustrative candidate at 0.80 K; the usual 0.50 K limit rejects the series.

TopSpin: three ramps with configurable Δ

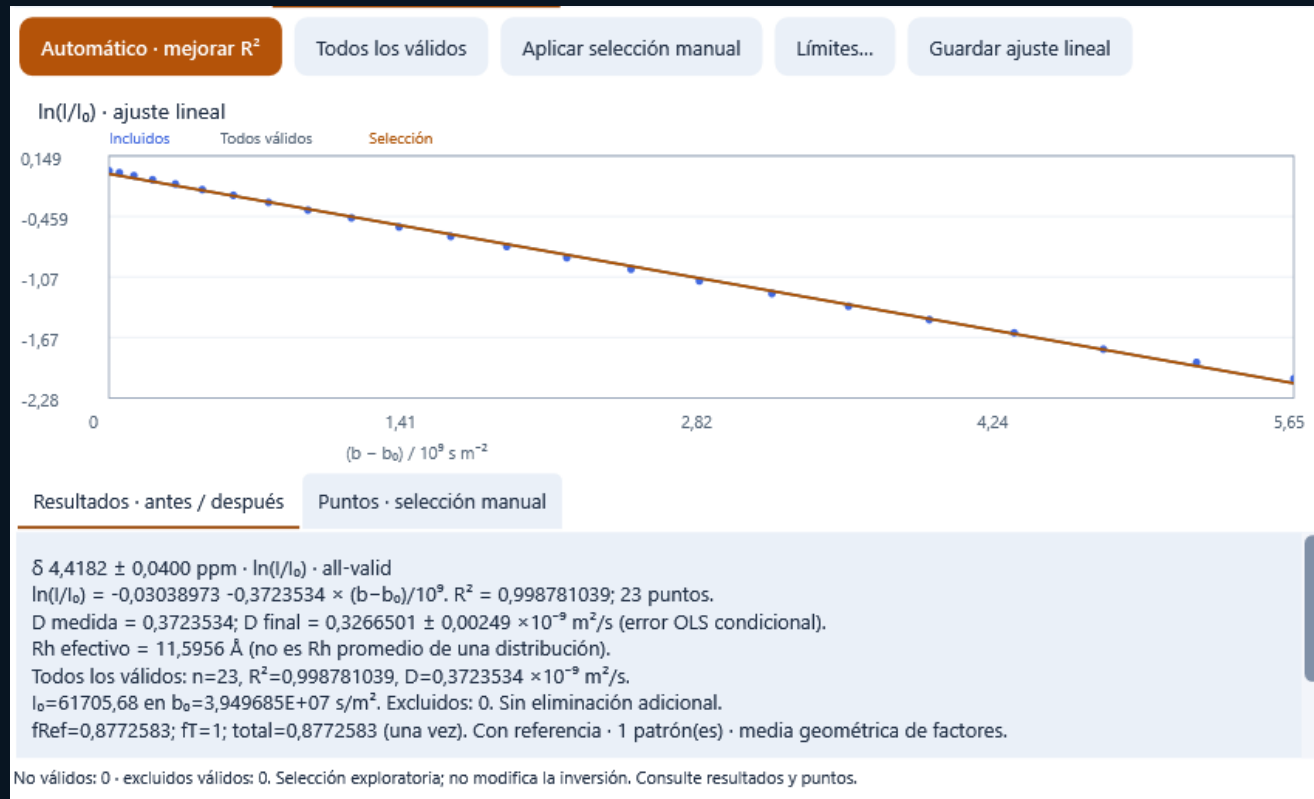
xpy dosy_workshop.py

Configurable example	D20 / s	EXPNO
Δ 50 ms	0.050	100–122
Δ 100 ms	0.100	200–222
Δ 150 ms	0.150	300–322

Same template · fixed P30 · complete gradient list · integrate the standard

The assistant prepares parameters and calibrates from the 1D series. Check destinations, units and limits before acquisition.

Prime: fit a real signal before the map



1 Open the series

2 Fix the region

3 Fit and review

D final = 0.3267

Result saved with its points, calibration and corrections

TP194 · 4.4182 $\pm$ 0.04 ppm · 23 points, 0 excluded · D / 10⁻⁹ m² s⁻¹, conditional on calibration and model.

Prime and ResinAtOnce: consistency between signal and map

Ramp / EXPNO	Local peak D	Map peak D	Shape difference
10:32	0.3270	0.3270	0.0 %
33:55	0.3426	0.3426	0.0 %
56:78	0.3806	0.3806	0.0 %

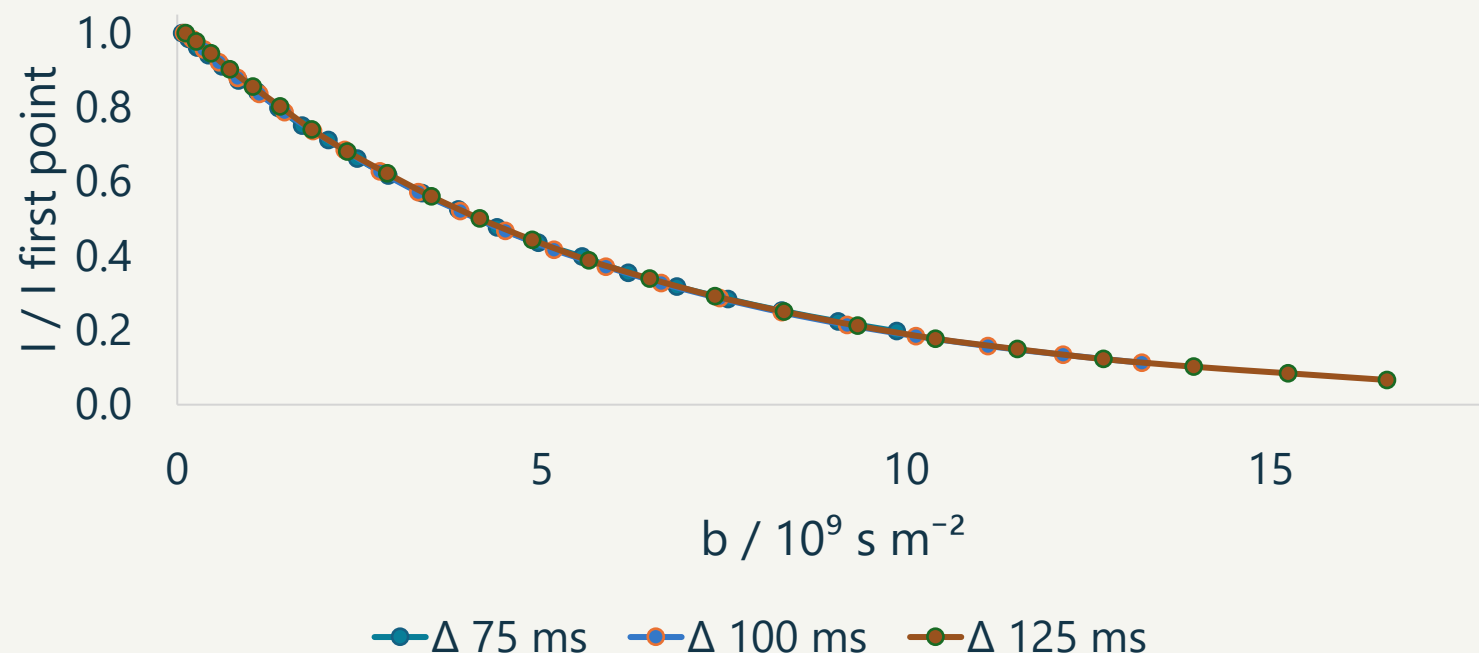
TP194 · ≈ 7.736 ppm · 2048-point D grid

Before: 84.87% shape difference between the local TRAln signal and the MF map

The same extraction, grid, algorithm and calibration in both paths

D / $10^{-9} \text{ m}^2 \text{ s}^{-1}$ · internal check on real data. Agreement does not establish species identity.

DALTAIL: a reproducible KK2 batch



- 1 Read FIDs and parameters
- 2 Process each block consistently
- 3 Extract and export decays

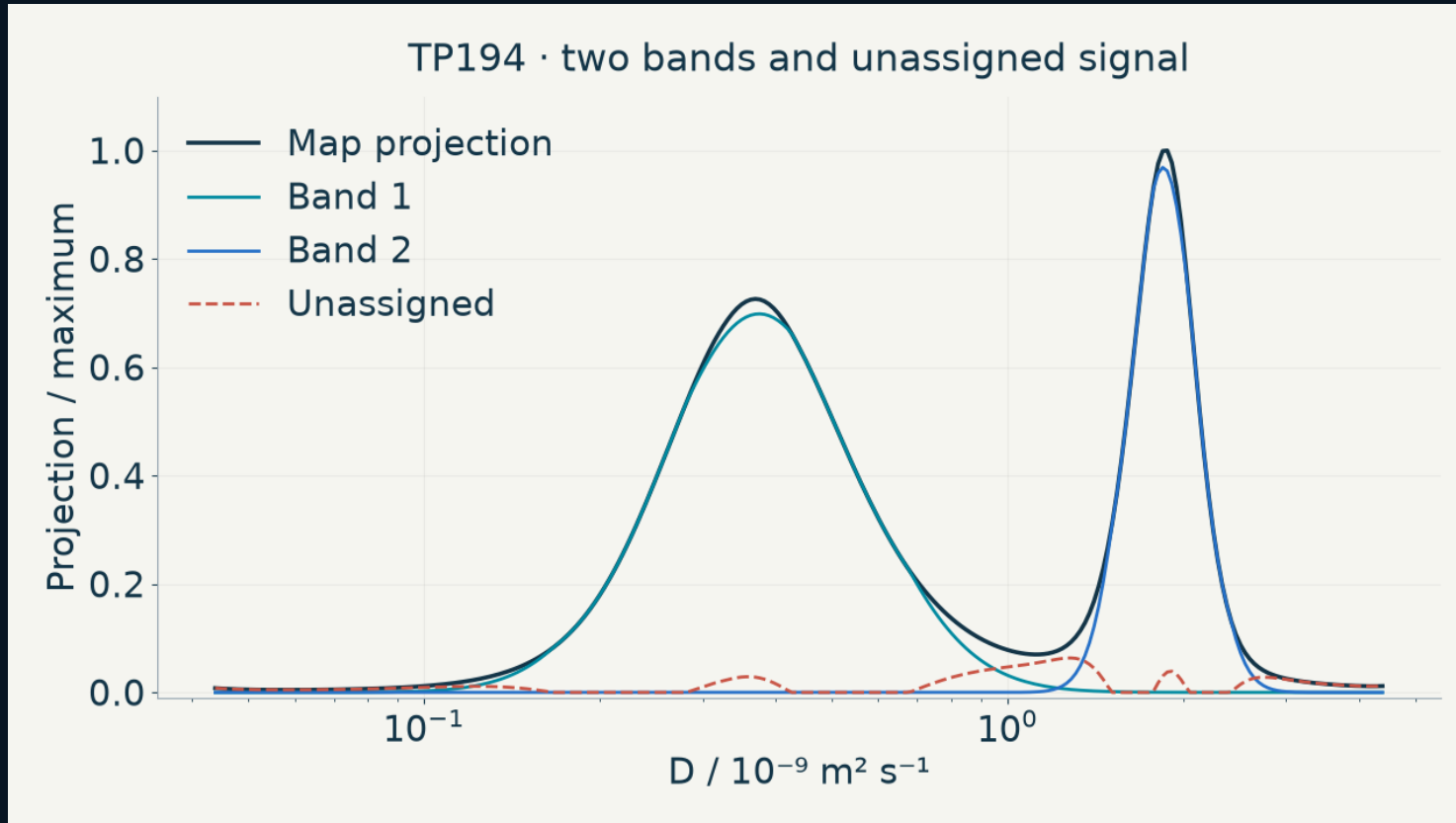
KK2-50 · OCH₃
3.45–3.70 ppm

Batch: 230 spectra · 10 ramps

DALTAIL repeats the analysis from a saved configuration

Archived decays. Primary D incorporates TTMS and temperature together; molecular masses remain model-dependent.

TP194 case · keep what the model cannot explain visible



**≈6%
unassigned**

Retain the signal
Compare methods
Hold out gradients

A useful map lets us review the fit and what remains unresolved

Real TP194 · archived projection case. Predictive checks are a subsequent Scientific Workbench step.

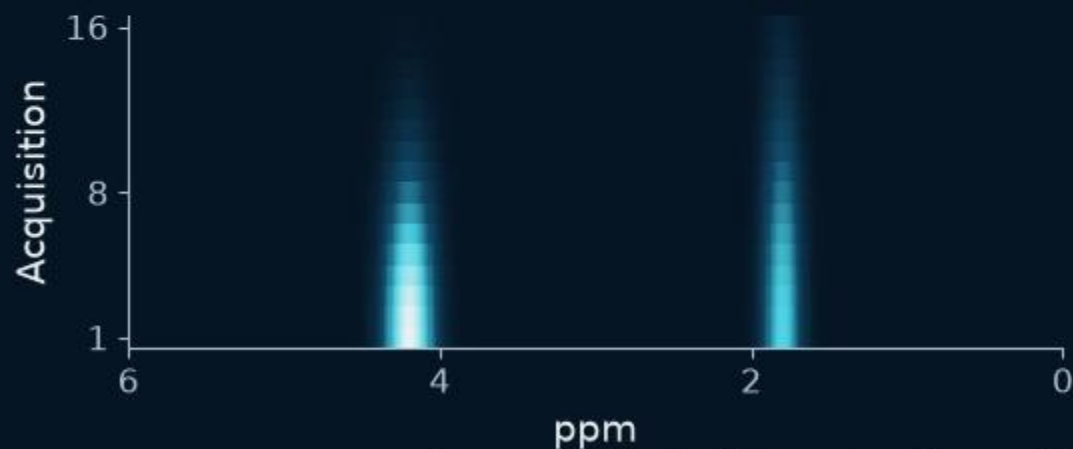
From repeating gradients to encoding space

SPEN-DOSY / ultrafast DOSY

ONE ACQUISITION PER GRADIENT



Build the series point by point

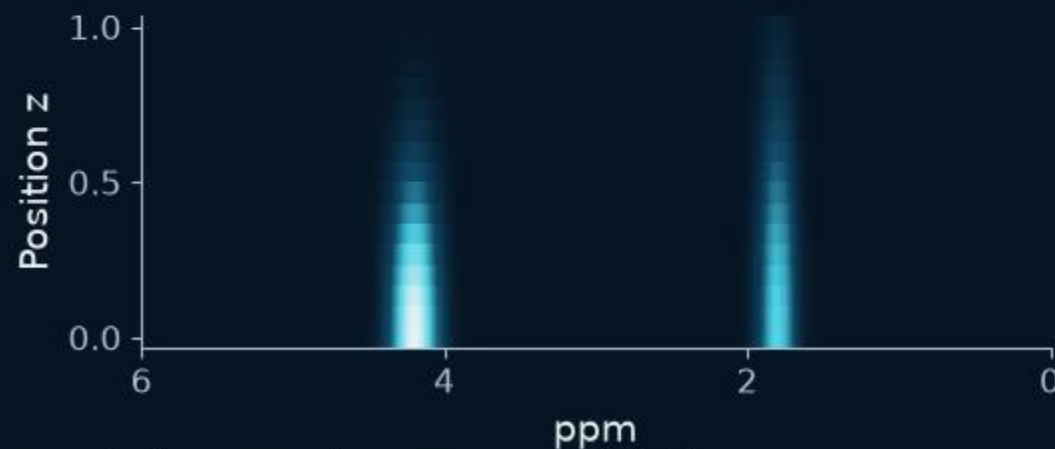


A parallel dimension; the $b(z)$ model remains essential

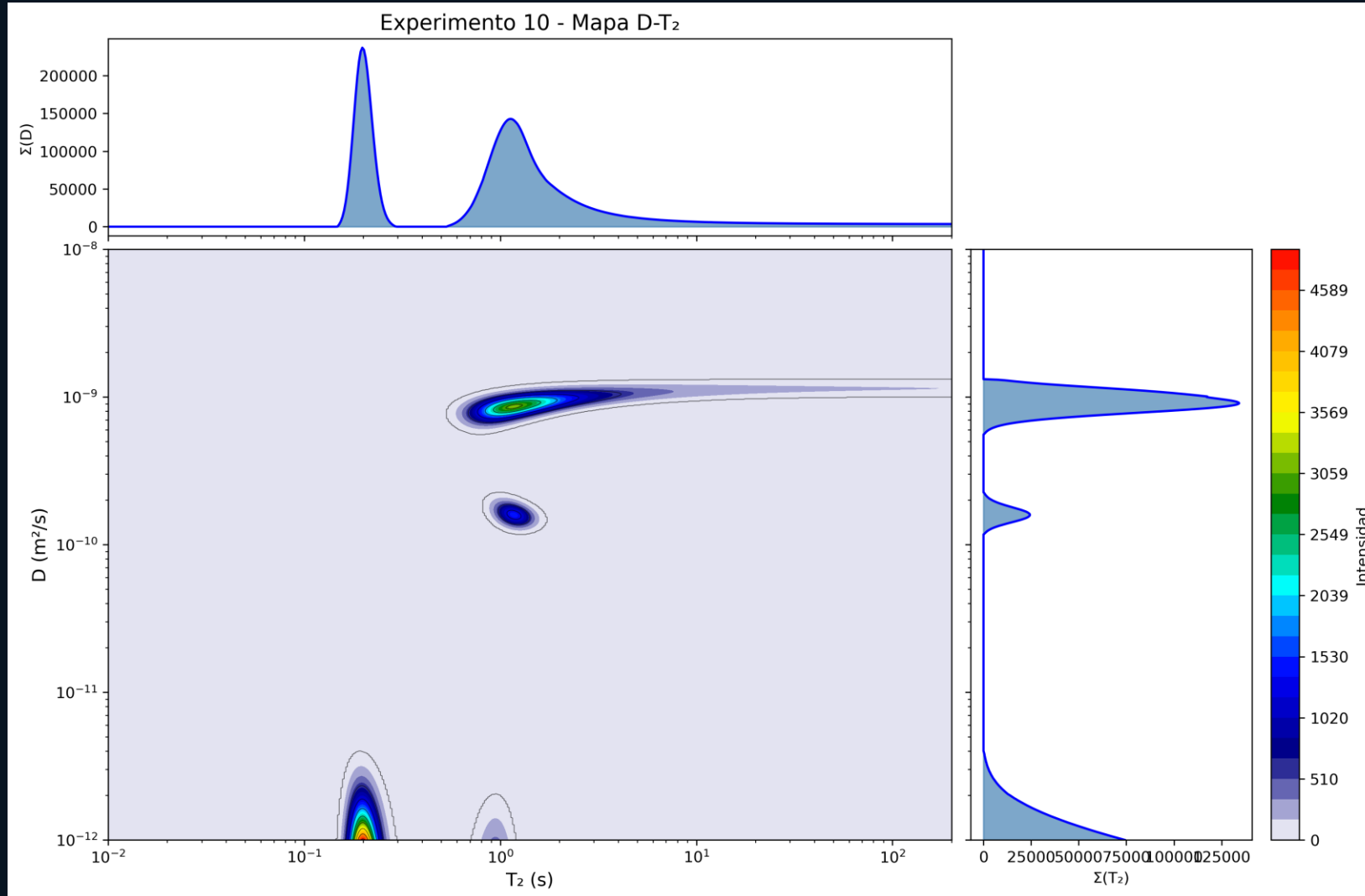
DIFFUSION DISTRIBUTED ALONG z



Recover all positions



Ultrafast: diffusion and relaxation in one map



UF D-T₂

Experiment 10
4 scans · 32 echoes
≈28 seconds

CURRENT DEVELOPMENT

**z gradient
+ EPSI readout**

**Validate $b(z)$
and sensitivity**

A carefully acquired signal.
A reproducible analysis.
A result we can check.

Calibrate and retain useful information

Automate repetitive steps

Review the result with examples and controls

Prime · local ILT 1/4

Method / component	What it is used for	Main caution
NNLS	Positive baseline without smoothing; diagnose minimum residual.	SciPy NNLS / active-set NNLS; explicit dictionary
Tikhonov	Continuous distributions with controlled smoothness; polymers/polydispersity.	Fixed λ ; check penalty order and grid scaling.
FLINT quadratic objective	Positive L2 objective for stable inversion; this library uses NNLS.	Same positive L2 objective; NNLS engine, not FLINT original iteration
GNAT ILT NNLS	Compare GNAT positive ILT branch at a manual lambda.	Manual lambda squared; excludes GNAT GCV/L-curve/UI peak picking
TRAIIn-style trust region	Iterative positive trust-region inversion; control its stopping criterion.	Prime uses termFac $\times$ NNLS residual; do not interchange it with noise radii.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Prime · local ILT 2/4

Method / component	What it is used for	Main caution
ITAMeD positive FISTA	Sparse distributions with few peaks through L1 and FISTA.	Fixed grid; check objective scaling before comparing lambda.
Positive smooth sparse inversion	Compromise between few peaks and smoothness.	L1 plus finite-difference quadratic
Positive maximum entropy	Positive distributions regularized by entropy relative to a reference.	Reference depends on data; check convergence and implementation.
PALMA entropy/L1 PPXA+	Combine peaks and broad distributions with entropy/L1 and a noise radius.	First observation must be positive; verify final feasibility.
Local curvature reweighted Tikhonov	Adapt local smoothing to curvature; compare sensitivity.	Not authentic UPEN2D/MUPEN2D

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Prime · local ILT 3/4

Method / component	What it is used for	Main caution
BRD dual positive L2	Solve positive L2 via a dual formulation; explicit lambda.	Fixed alpha; no OSILAP automatic selection or compression.
Chambolle-Pock positive L1	Solve sparse L1 inversion by primal-dual splitting.	Not PDHGM2; inspect the projected-gradient residual.
Regularized block Kaczmarz	Solve regularized inversion using row/block updates.	Row order matters; the grid adapter uses internal defaults.
Positive L1-LS	Positive L1 solution by interior point; optimization reference.	Fixed lambda and dense Newton; not the original large-scale PCG.
FLINT-FISTA	Iteratively solve the positive FLINT quadratic objective.	Step and stopping differ from upstream; check convergence.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Prime · shared structure

Method / component	What it is used for	Main caution
Positive row-group sparse inversion	Shared D support across multiple frequencies.	Shared support is not molecular identity; one column reduces to L1.
Nonnegative sparse nuclear regularization	Joint map with few profiles and sparse significant coefficients.	ADMM; not branded ADSpLRU/LRSpILT
Reweighted nuclear heuristic	Strengthen rank/support selection through adaptive weights.	Changing weights; no fixed convex objective certificate
Orthogonal matching pursuit dictionary adapter	Select few D atoms from a dictionary; explicit support limit.	Column scaling affects selection; no automatic noise estimate.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Prime · joint maps 1/2

Method / component	What it is used for	Main caution
MF-NNLS	Joint distribution map with shared factors.	Predictive rank $\neq$ molecules; inspect λ_S , λ_A , bounds and residuals.
TRAI _n -MF	Joint model with signed attenuations; explicit legacy replay.	Historical identifier label does not describe the revised route. Not per-bin TRAI _n .
Per-bin TRAI _n	Same local fit for every frequency in a region.	Does not impose inter-frequency correlations or shared factors.
RAI-S	Few discrete D values with spectral support selection.	Not RAI-Net or a trained network; a broad distribution may violate the discrete model.
DOMES	Shared discrete rates with coverage and support review.	Check boundaries, held-out prediction and component count.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Prime · joint maps and 2D

Method / component	What it is used for	Main caution
SILT-DOSY	Joint map with sparse and low-rank penalties.	Nonconvex reweighting; operational stop $\neq$ global optimum.
SILT original ADSpLRU	Replay the supplied original core with diagnostics.	Historical stop $\neq$ KKT certificate; separate from revised SILT.
TRAI _n 2D · C#	Separable D–T ₂ inversion or other two-operator problems.	No lambda; termFac and NNLS residual. Physical kernel/profile must be reviewed.
FISTA 2D · C#	Positive L1 2D inversion using a separable operator.	Not a verbatim ITAMeD copy; KKT does not validate the physical model.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Library · discrete components 1/2

Method / component	What it is used for	Main caution
Positive variable projection	Fit a fixed number of exponential components without a D grid.	Fixed rank and local minima; no guaranteed identification.
SCORE	Separate component spectra by sharing D across frequencies.	Request signed amplitudes; the generic VP default is positive.
OUTSCORE	Unmixing that penalizes overlap between component spectra.	GNAT absolute spectral cross-talk objective; fixed rank
HYSORE	Balance reconstruction and spectral separation in a discrete model.	GNAT hybrid residual/cross-talk objective; explicit relative weight
AIC/BIC selection over VP	Choose model order among variable-projection models using AIC/BIC.	Not original Provencher DISCRETE or SPLMOD

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Library · discrete components 2/2

Method / component	What it is used for	Main caution
DECRA	Algebraic decomposition of exponential components with uniform b.	Requires uniform b steps and sufficient rank; poles may be nonphysical.
Explicit-region local SCORE	Apply SCORE to explicitly selected ppm regions.	Use disjoint regions; segmentation is not automatic.
Automatic LOCODOSY local inversion	Segment regions, estimate local order and reduce components.	Adapted segmentation; reported intervals are not calibrated uncertainty.
ESPIRA-II AAA-selected Loewner pencil	Identify exponential sums through poles and a fixed rank.	Uniform sampling and fixed rank; retain nonphysical poles.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Library · decomposition and FID

Method / component	What it is used for	Main caution
MCR-ALS	Multivariate decomposition with constraints and optional exponential profiles.	Rotational ambiguity remains; an exponential fit is needed to obtain D.
Symmetric FastICA	Explore statistically independent factors; fit D afterwards if appropriate.	Statistical independence estimates neither D nor molecular identity.
Three-way CP-ALS	Three-dimensional data with factors shared across modes.	Local solution; it does not perform DOSY inversion by itself.
GNAT Fourier RRT	Fourier-Laplace reconstruction from complex FID and uniform b.	Requires uniform b and correctly oriented FID; output is a complex map.
GNAT Fourier FDM	Resolve poles/frequencies from FIDs through filter diagonalization.	Numerical policies differ; output is not a calibrated physical density.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Library · linear cores

Method / component	What it is used for	Main caution
Signed L1 interior point	Signed sparse linear fits; not automatically a positive distribution.	Signed output does not represent positive masses without constraints.
PLSS residual recursive solver	Consistent linear systems; auxiliary unregularized core.	For consistent systems; signed, unregularized output.
IRLS (fixed p) dictionary adapter	Explore sparse solutions through reweighted least squares.	Signed output; iterate stability does not prove global optimality.
Weighted PLSS recursive core	Weighted consistent linear systems; auxiliary core.	Consistent systems and signed output; not every PLSS variant is included.
CP-ALS 3D / 4D	Factorize three/four-way tensors; multi-experiment studies.	Local solution; stability is not a stationarity certificate.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Library · other Laplace cores

Method / component	What it is used for	Main caution
ADSpLRU archived reweighted ADMM	Joint sparse/low-rank unmixing with reweighted ADMM.	Historical residual stop is not a fixed convex- objective certificate.
IPSpLRU incremental proximal recurrence	Joint unmixing with an incremental proximal recurrence.	Explicit initialization and update count; convergence is not guaranteed.
LRSpILT archived ADMM recurrence	Joint DOSY reconstruction using low-rank and sparse structure.	Adaptive thresholds; relative stopping is not an optimality certificate.
PDHGM2 NMRIversions.jl	Replay the NMRIversions.jl recurrence and check KKT separately.	Projection does not guarantee the positive objective; inspect actual KKT.
ILT.jl positive ridge	Positive inversion with a signed baseline term.	NNLS replaces lpopt; baseline is also penalized and alpha is fixed.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Library · regularization and unmixing

Method / component	What it is used for	Main caution
TailoredNorm archived p sweep	Explore a p-norm sweep; replay the archived numerical policy.	The original threshold can annihilate output; retain this failure.
UPEN2D adaptive spatial penalties	2D distributions with adaptive local smoothing; two physical operators.	NNLS inner solver; no original compression. Check both operators.
MUPEN2D spatial multi-penalty FISTA	2D inversion with adaptive multiple penalties; signed output.	Signed output; distinguish corrected adjoint from the historical policy.
SUnSAL unmixing ADMM	Sparse unmixing with an explicit dictionary and optional constraints.	Justify positivity and sum-one; L1 is constant on a positive simplex.
CLSUnSAL collaborative unmixing	Collaborative unmixing with support shared across columns.	Historical stopping is not an optimality certificate.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Library · specific domains

Method / component	What it is used for	Main caution
Fully constrained least squares	Positive sum-to-one unmixing only when normalization justifies it.	Sum-one is mandatory; justify it for DOSY intensities.
MRI mixture of Wisharts/tensors matrix core	Directional MRI diffusion using Wishart/tensor mixtures.	Not scalar DOSY ILT; directional encoding is required.
MRI Q-ball harmonic transform	Angular Q-ball MRI reconstruction in a supplied harmonic basis.	External harmonic basis; neither scalar ILT nor normalized probability.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Context · models and engines

Method / component	What it is used for	Main caution
Mono / multi-exponential fitting	Few expected components; start with monoexponential fitting on an isolated signal.	A k-exponential fit assumes k; nearby D values are hard to resolve.
CONTIN	Positive continuous distribution with regularization and its own parameter selection.	Fixed-lambda Tikhonov is not the full CONTIN program.
SVD / TSVD	Diagnose informative rank; compress the problem or truncate weak modes.	SVD alone does not guarantee positivity; more bins do not create resolution.
FISTA	Numerical engine for L1 or L2 objectives with projection/proximal operators.	The engine alone defines neither the prior nor the experiment.
UPEN / 2DUPEN / MUPEN2D	Adaptive local smoothing for sharp peaks and broad regions.	Specify version, positivity and operators; not generic curvature_reweighted.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Context · multidimensional and learning

Method / component	What it is used for	Main caution
EDMILT	Multidimensional ILT with tailored regularization; D–T2/T1–T2 correlations.	Requires multidimensional data/operators; not assumed implemented in Prime selector.
DRECT	Learned Laplace-map reconstruction from synthetic training.	Check kernel, SNR and domain; a checkpoint is not local validation.
DREAM	Learned reconstruction with aleatoric uncertainty estimation.	Network uncertainty certifies neither identity nor coverage outside training.
CoMeF / DOSY-Net / DLEMLR / DILT	Other learned architectures for ILT/DOSY/multidimensional data.	Check architecture, weights and domain; archived code need not be in Prime.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Context · discrete and selection

Method / component	What it is used for	Main caution
DISCRETE / SPLMOD / Prony	Finite exponential sums; poles and amplitudes.	Discrete assumption, order and sampling matter; not a continuous distribution.
Landweber / CG / LSQR	Iterative engines; early stopping as regularization.	Residual convergence does not prove component resolution.
Discrepancy / L-curve / GCV	Choose regularization or rank and study stability/uncertainty.	Selection/diagnostic strategies; not new solvers or experimental truth.

Reference catalogue · details, availability and sources in notes and accompanying CSV / TXT

Sequence-dependent equations

$$b = (\gamma g \delta)^2 F$$

Rectangular gradients	F
PGSE / PGSTE	$\Delta - \delta/3$
Balanced bipolar	$\Delta - \delta/3 - \tau/2$
Double bipolar	$\Delta - 2\delta/3 - (\tau_1 + \tau_2)/2$

Bipolar δ : sum of lobes · τ : internal gap including RF

Double: two equal $\Delta/2$ blocks; verify the program definition

stebpgp1s1d + SMSQ10.100

$$b = 0.81 (\gamma g \delta)^2 [\Delta - 0.32525 \delta - \tau/2]$$

$$\delta = 2 \times P30$$

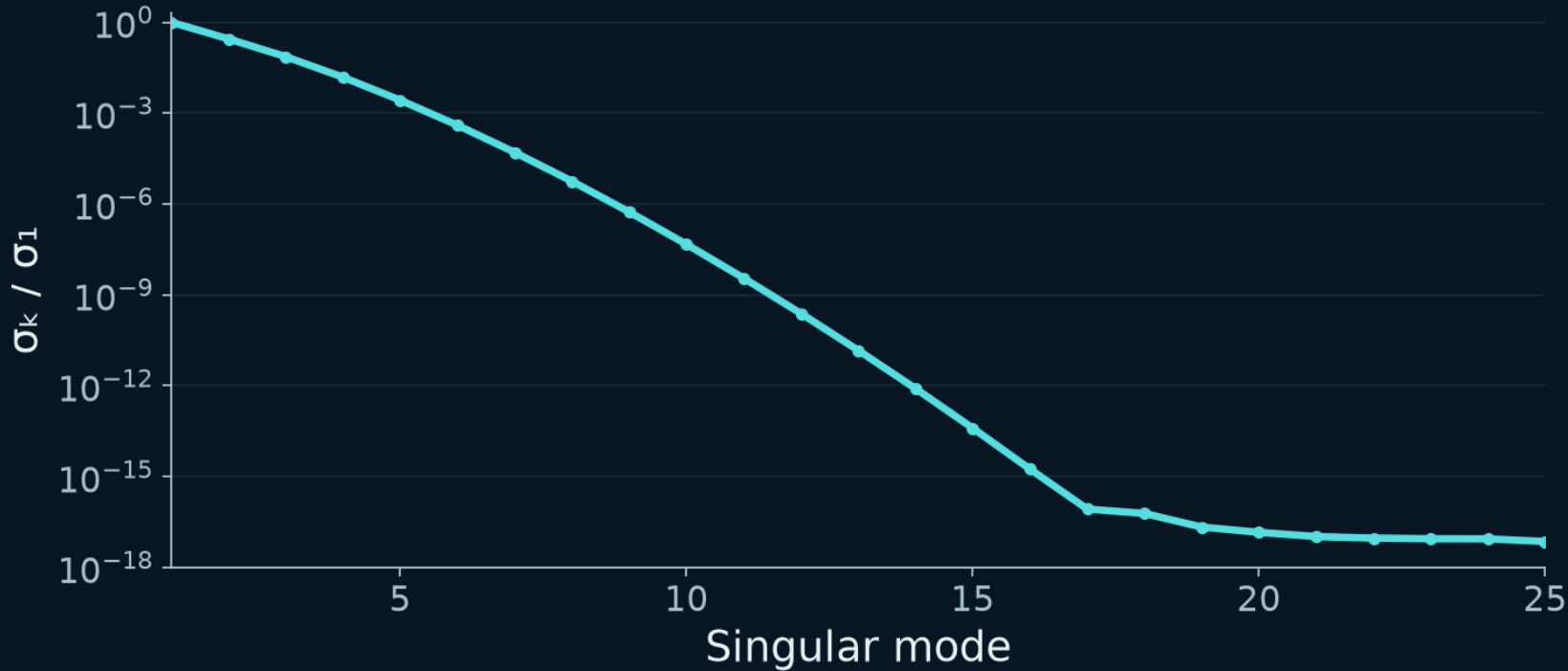
Area = 0.90
In b: $0.90^2 = 0.81$

$$\tau = D16 + P2$$

The TBO profile uses $\tau \approx D16$
Its measured calibration is retained

Shape, timing, G and Dref(T) must correspond to the same experimental profile.

Noise is amplified when the decay is inverted



$$\mathbf{S} = \mathbf{K}\mathbf{f} + \boldsymbol{\varepsilon}$$

25

measurements

110

weights to reconstruct

Regularization imposes a preference on the solution

SIMULATION · weak kernel directions · small residuals admit very different solutions

Each algorithm favours a different solution

Observed problem	Family / examples	Preference and risk
Isolated signal	Monoexp. / NNLS	Minimal model / unstable unregularized peaks
Broad distribution	Tikhonov / MaxEnt / PALMA	Regularization / may merge components
Few components	ITAMeD / L1 / VP / SCORE	Sparsity or model order / wrong assumptions
Overlapping signals	MF / TRAIIn-MF / SILT	Shared structure / spurious factors

Compare solutions with the same **b** values, noise and preprocessing

Complete catalogue in the appendix: uses, limits and availability. FISTA/ADMM/SVD are engines, not physical models.